EXPERIMENT REPORT

Practical Record – Computer Science and Business Systems

Automating business processes using Deluge scripting and workflows with application testing and debugging

Bank Loan Application Form

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1. Aim

To design and implement a workflow automation system using Zoho Creator for a Bank Loan Application Form. The main objective is to automatically send an acknowledgement email to the applicant whenever a loan application is submitted successfully. This automation helps confirm the receipt of the application, improves communication between the applicant and the loan processing team, and reduces the need for manual email communication.

2. Software Requirements

- **Platform:** Zoho Creator
- **Browser:** Google Chrome
- **Application:** Bank Loan Application Form
- **Module:** Workflow Automation
- **Action:** Email Notification
- **Email Service:** Gmail or another supported email service

3. Introduction

Workflow automation allows an application to perform routine tasks automatically when a defined event occurs, without manual intervention. In a loan application process, an applicant submits personal and financial details through an online form, and the organization needs to confirm to the applicant that the submission has been received. Handling this confirmation manually is time-consuming and can cause delays.

In Zoho Creator, a workflow is attached to a form and is triggered by a form event. In this experiment, the workflow is triggered by the Created record event together with the Successful form submission form event. As soon as a new loan application record is saved, the workflow executes its configured action. Deluge is the scripting language of Zoho Creator and can be used for custom logic, although in this experiment the required behaviour is achieved with the built-in Notification – Send Email action.

An acknowledgement email improves communication with the applicant by confirming immediately that the application has been received and by informing the applicant that the loan processing team will review it and make contact regarding the next steps. Because the recipient is taken from the Email Address field of the submitted record, every applicant receives the message at the address they provided.

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4. Procedure

Step 1: Selecting the Workflow Option

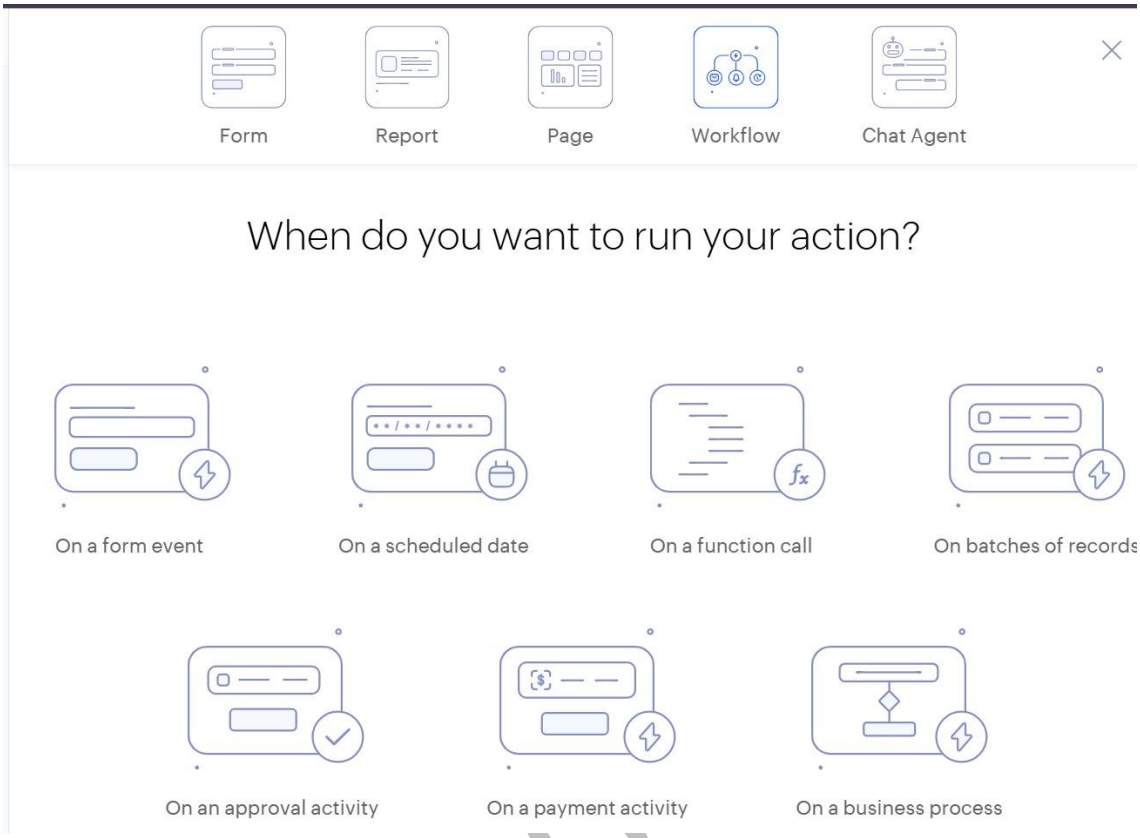


Fig. 1: Workflow option and available execution methods in Zoho Creator

Description

Initially, the Bank Loan Application application is opened in Zoho Creator. The Workflow option is selected to automate the actions associated with the loan application process. Zoho Creator provides different workflow execution methods, such as form events, scheduled dates, function calls, batches of records, approval activities, payment activities, and business processes.

For this experiment, the form event option is selected because the email notification must be triggered when an applicant submits the loan application form. This ensures that the workflow responds automatically to the applicant's submission instead of requiring the user to perform the notification manually.

Step 2: Creating and Configuring the Workflow

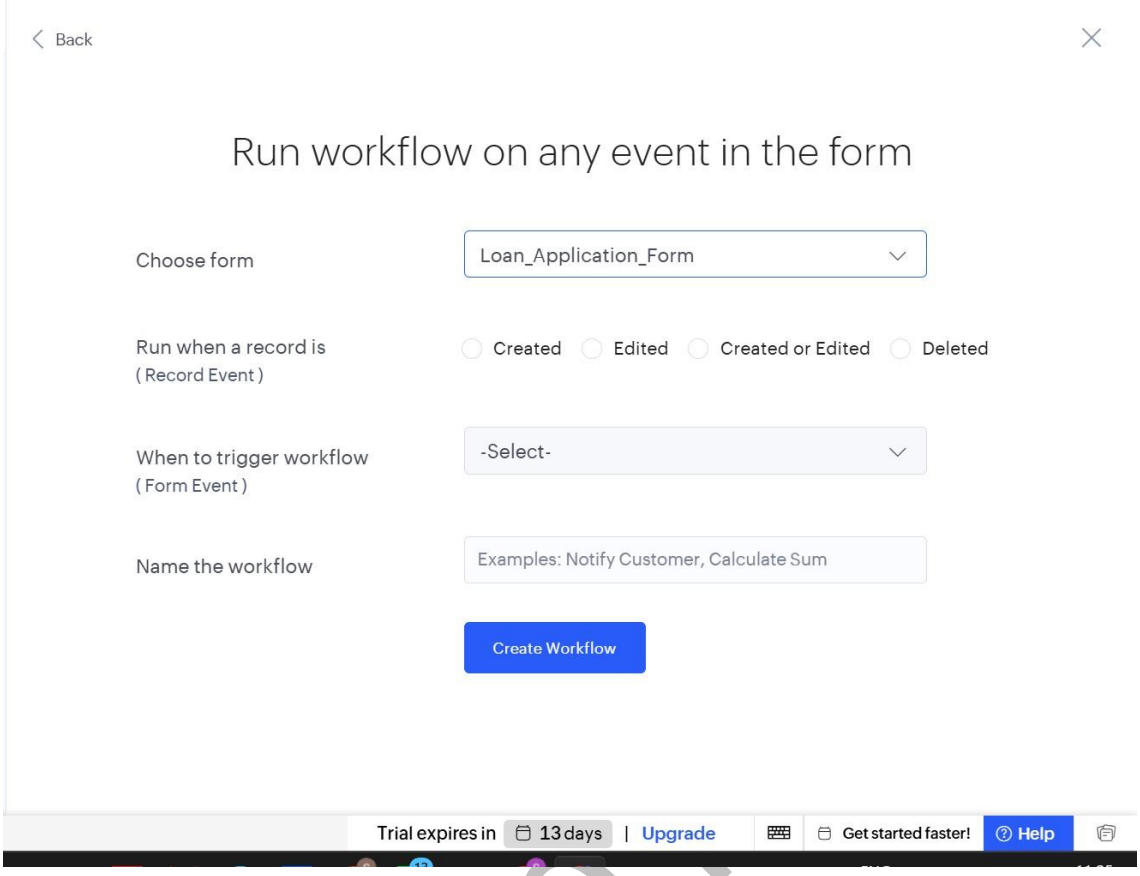


Fig. 2: Workflow configuration – form, record event, form event and workflow name

Description

After selecting the form event, the workflow configuration screen is displayed. The Loan_Application_Form is selected from the Choose Form dropdown menu. Under the record event section, the Created option is selected because the workflow should execute when a new loan application record is created.

Next, the form event is configured as Successful form submission. This setting ensures that the workflow runs after the form has been submitted successfully. The workflow is named Loan_Application_Notification, which clearly identifies its purpose. Finally, the Create Workflow button is clicked to create the workflow rule in Zoho Creator.

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Step 3: Adding an Action to the Workflow

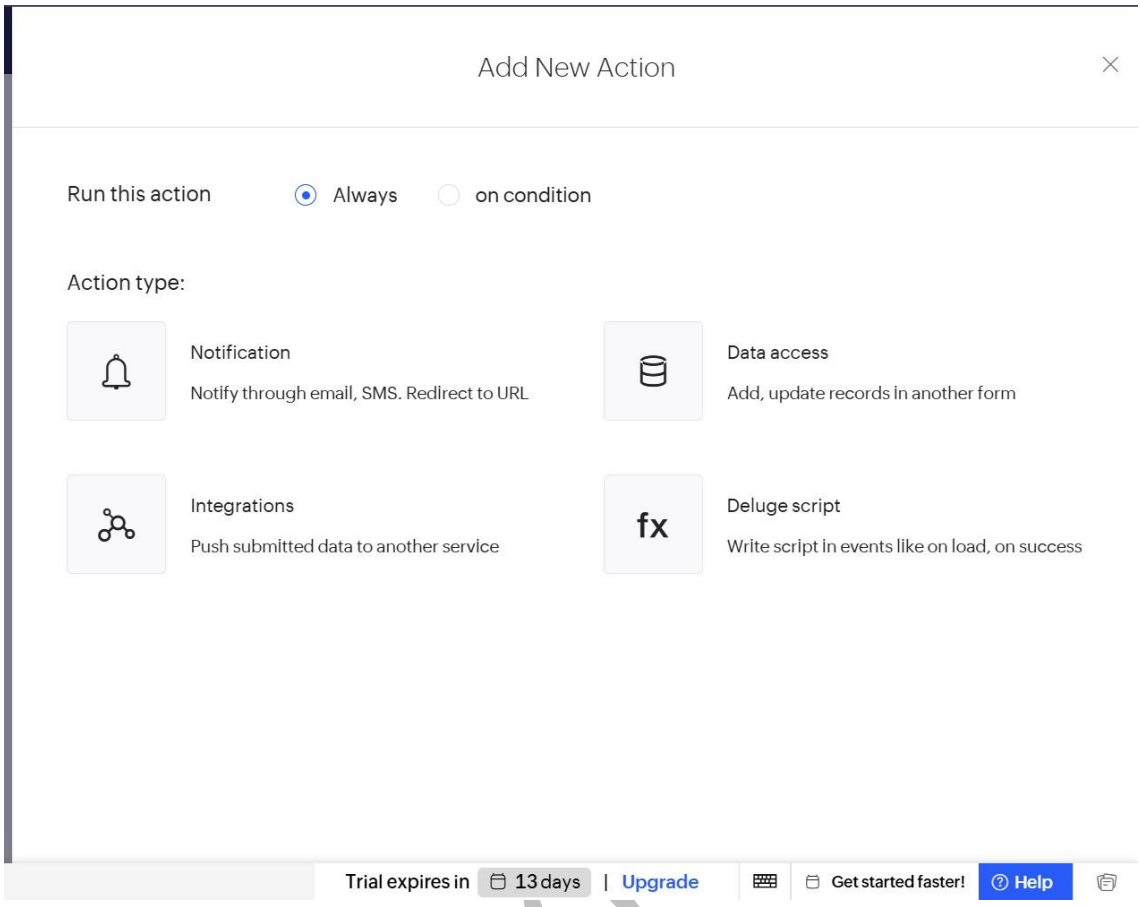


Fig. 3: Action configuration screen with the Add New Action option

Description

After creating the workflow, the action configuration screen is opened to specify what should happen when the workflow is triggered. The Add New Action option is selected to add an automated operation to the workflow.

Zoho Creator provides several action types, including Notification, Data Access, Integrations, and Deluge Script. These actions allow users to send emails, manage records, connect with external services, or execute custom business logic.

In this experiment, the Notification action type is selected because the primary requirement is to send an acknowledgement email to the applicant after successful loan application submission. The action is configured to run whenever the workflow conditions are satisfied.

Step 4: Selecting the Send Email Action

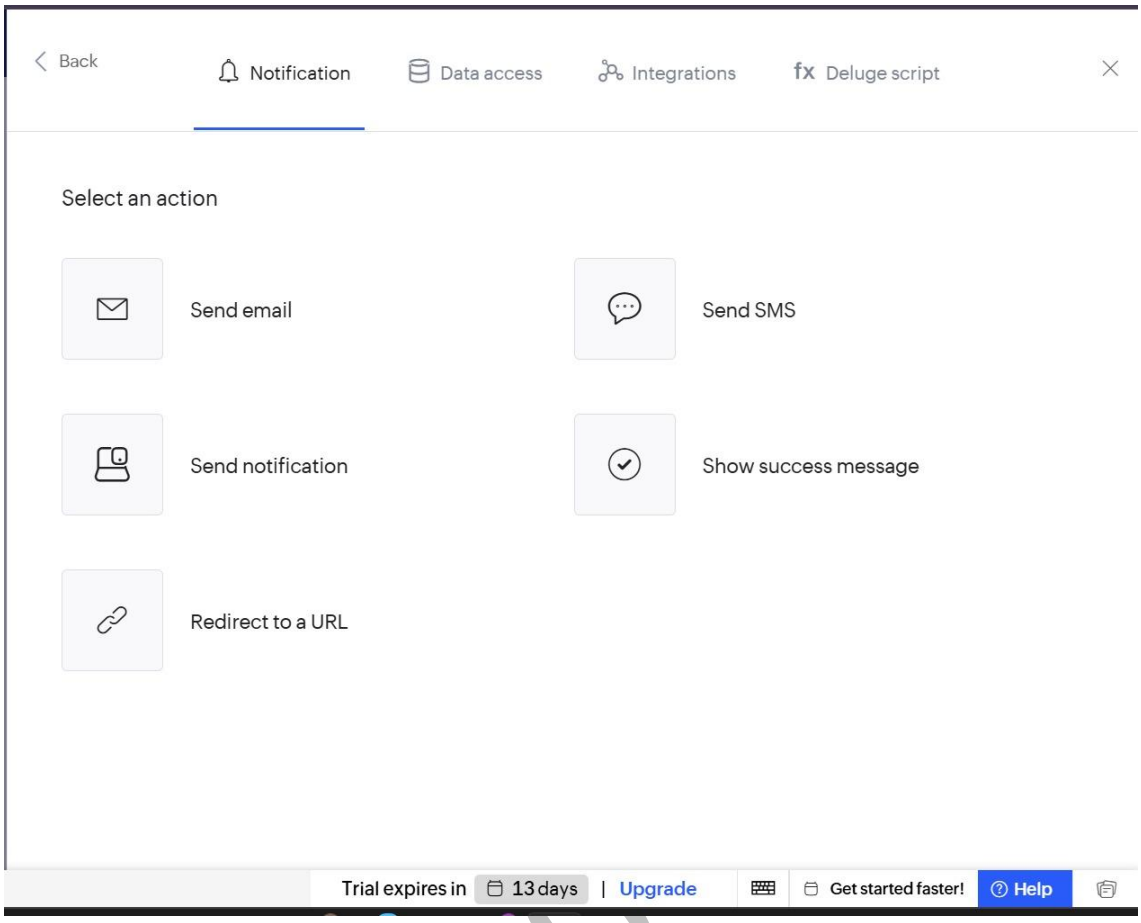


Fig. 4: Notification actions available in Zoho Creator, including Send Email

Description

After selecting the Notification category, the available notification actions are displayed. These include Send Email, Send SMS, Send Notification, Show Success Message, and Redirect to a URL.

The Send Email option is selected to establish an automatic email communication process between the application and the applicant. This action allows the developer to specify the recipient's email address, the subject of the message, and the email content.

The recipient is mapped to the Email Address field available in the Loan_Application_Form. Therefore, each applicant can receive an acknowledgement at the email address provided during form submission, rather than sending the notification to a fixed recipient.

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Step 5: Configuring the Email Notification

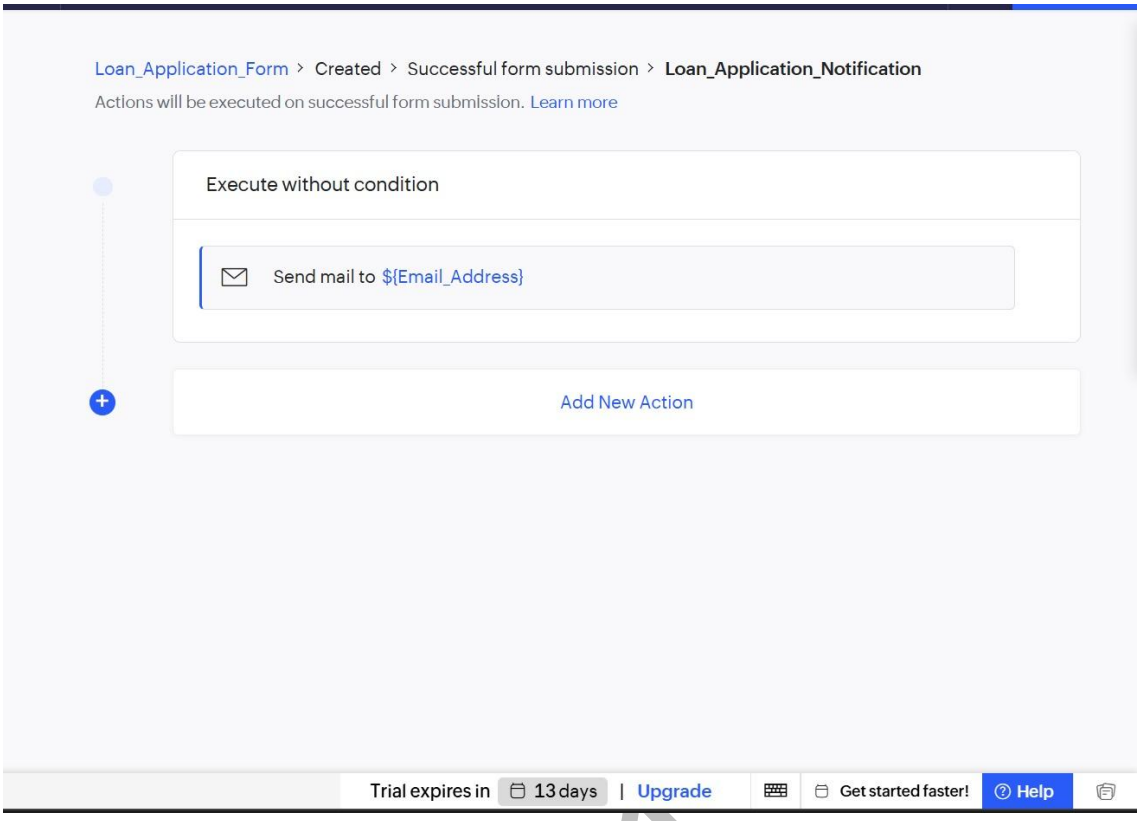


Fig. 5: Send Email action added to the workflow, addressed to ${Email_Address}

Description

After selecting the Send Email action, the email notification is configured with the necessary recipient details and message content. The applicant's email address is selected dynamically from the submitted loan application record using the Email_Address field.

The email subject is set to Loan Application Received Successfully to clearly communicate the purpose of the message. The body of the email contains a personalized greeting and an acknowledgement stating that the loan application has been received successfully.

The message also informs the applicant that the loan processing team will review the application and contact them regarding the next steps. This makes the communication process clear and professional. After entering the required details, the email action is saved and associated with the workflow.

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Step 6: Verifying the Workflow Status

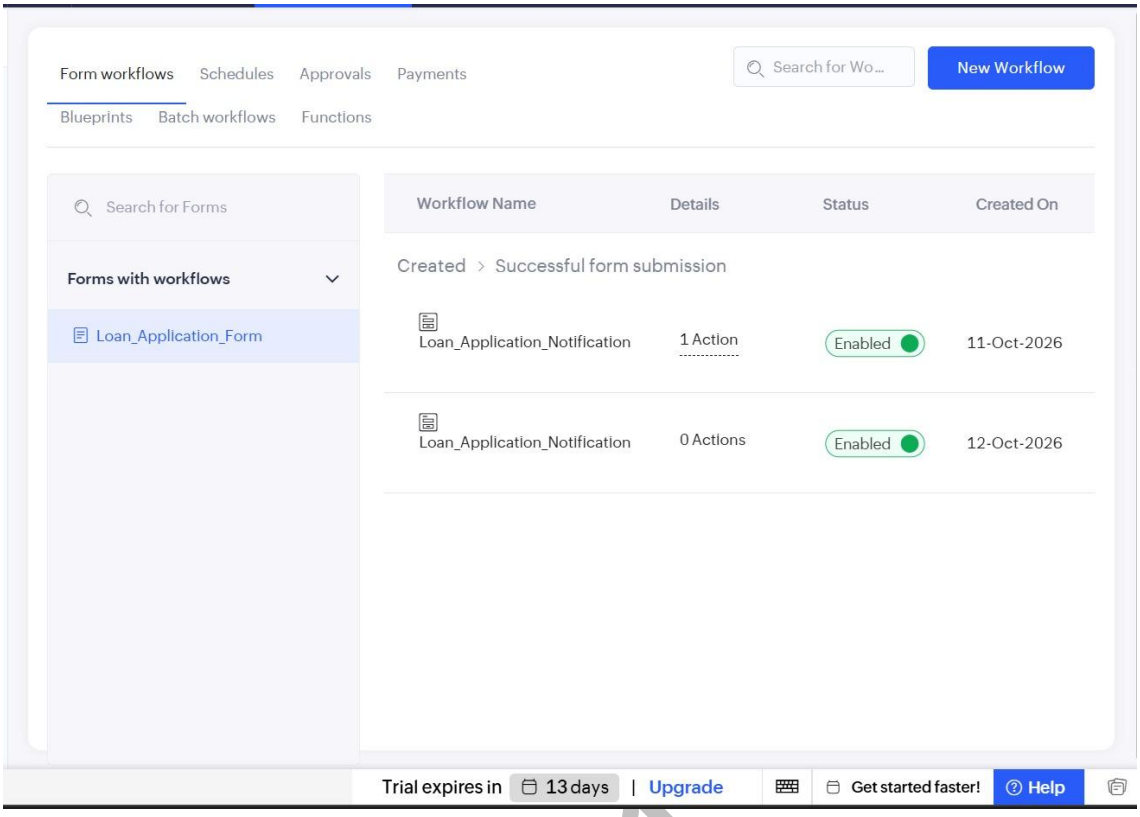


Fig. 6: Workflow list showing workflow name, action count, status and creation date

Description

After configuring the email action, the Workflow section is opened to verify the configuration. The workflow list displays the workflow name, the number of configured actions, the current status, and the creation date.

The Loan_Application_Notification workflow is checked to ensure that it is enabled and contains the Send Email action. An enabled workflow can execute its configured action when the specified event occurs.

This verification step is important because creating a workflow alone is not sufficient; the appropriate action must also be configured and saved. If multiple workflows have been created during testing, the correct workflow must be identified to avoid confusion or duplicate email notifications.

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Step 7: Submitting the Loan Application Form

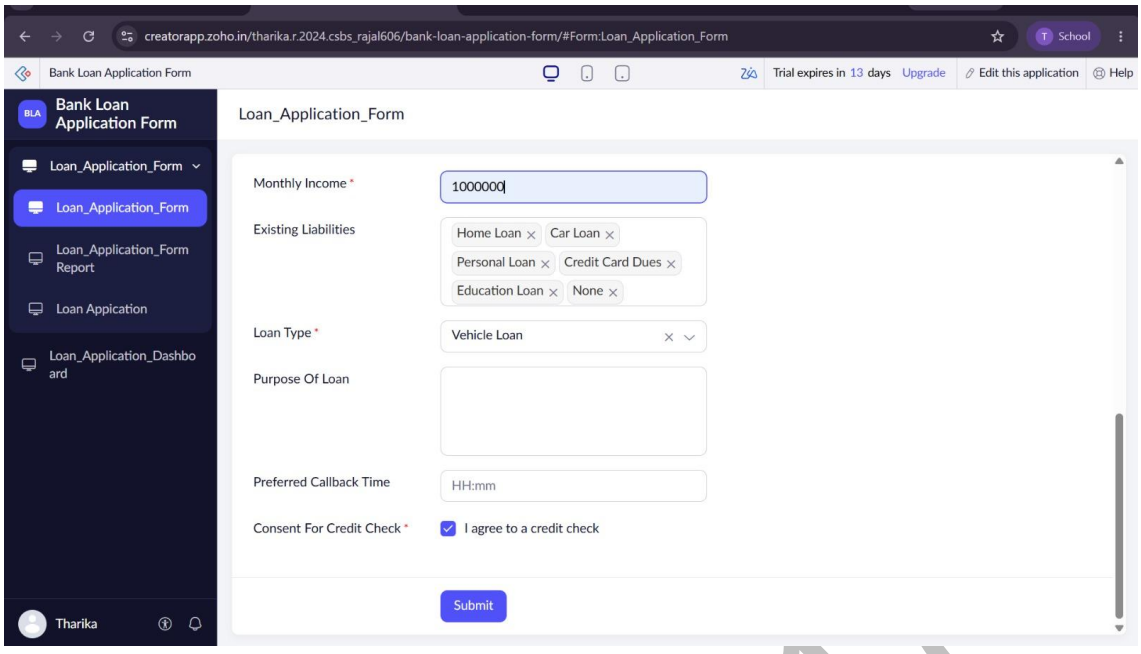


Fig. 7: Loan Application Form completed with sample applicant information

Description

The Bank Loan Application Form is opened to test the workflow with sample applicant information. The form contains various fields required to collect personal and financial details from the applicant.

The applicant enters the monthly income, selects existing liabilities, chooses the required loan type, and provides the purpose of the loan. The preferred callback time can also be entered to indicate a suitable time for communication. The applicant then provides the required consent for the credit check.

Before submission, all mandatory fields marked with an asterisk are completed, and the email address is checked for accuracy. The Submit button is clicked to save the application record. Once the submission is successful, the configured workflow should execute and initiate the email notification.

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Step 8: Verifying the Received Email

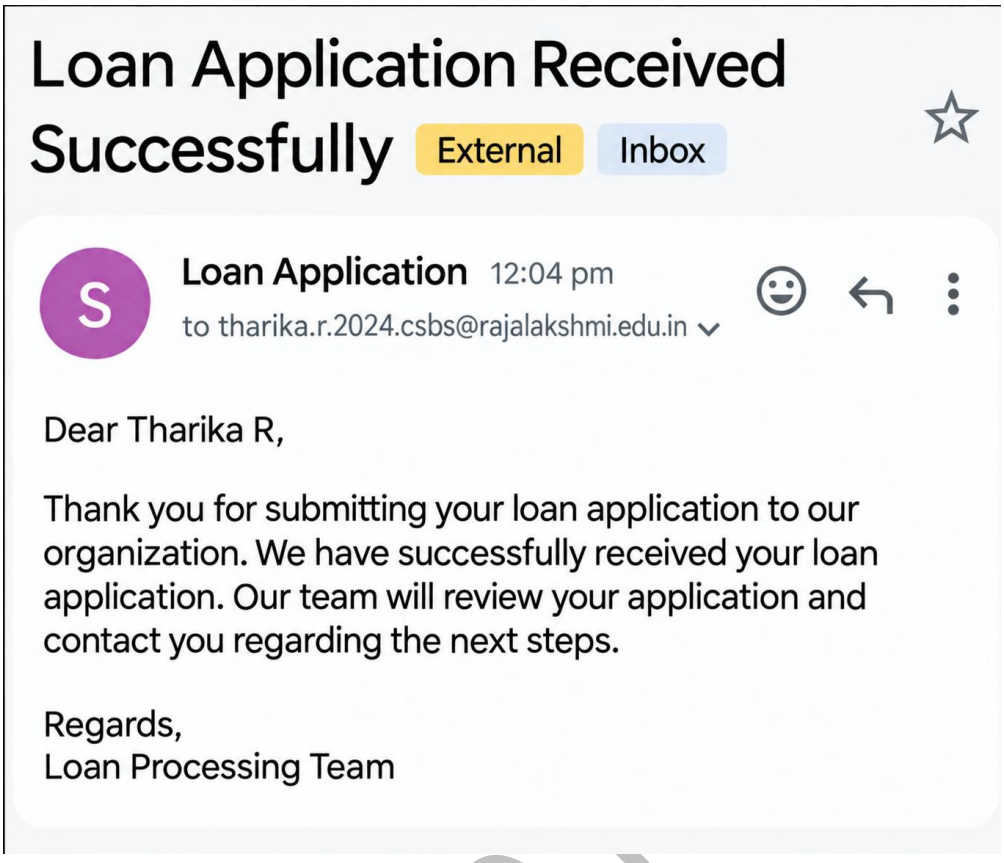


Fig. 8: Acknowledgement email received in the applicant's inbox

Description

After submitting the loan application form, the applicant's email inbox is checked to verify whether the automated acknowledgement message has been delivered. The received email displays the subject Loan Application Received Successfully and is addressed to the applicant's email address.

The message begins with a personalized greeting to Sharmila G. and confirms that the loan application has been received successfully. It further explains that the loan processing team will review the application and contact the applicant regarding the next steps.

This screenshot serves as evidence of the expected output of the workflow. It demonstrates the intended communication between Zoho Creator and the applicant's email account. When the message is received after form submission, it confirms that the email notification workflow is functioning as expected.

5. Workflow Configuration Summary

The table below summarizes the workflow configuration, as shown in Figures 2, 4, 5 and 6.

Configuration	Value
Form	Loan_Application_Form
Record Event	Created
Form Event	Successful form submission
Workflow Name	Loan_Application_Notification
Action Type	Notification – Send Email
Recipient	Applicant's Email Address field (${Email_Address})
Status	Enabled

Note: Figure 6 shows two workflows with the same name, Loan_Application_Notification, both enabled under Created > Successful form submission. The entry created on 11-Oct-2026 contains 1 action (the Send Email action), while the entry created on 12-Oct-2026 contains 0 actions. The workflow with the configured Send Email action is the one that sends the acknowledgement email.

6. Email Notification Example

The panel below shows the configured subject and message text of the acknowledgement email. It is a formatted example of the template used in the experiment; the actual received email is shown separately in Figure 8.

SAMPLE EMAIL (template text)
To: tharika.r.2024.csbs@rajalakshmi.edu.in
Subject: Loan Application Received Successfully

Dear Tharika R.,

Thank you for submitting your loan application to our organization. We have successfully received your loan application. Our team will review your application and contact you regarding the next steps.

Regards,
Loan Processing Team

7. Testing and Verification

The workflow was tested by submitting the Loan Application Form with sample applicant information and then checking the applicant's inbox. Each check below is marked according to what the screenshots in this report actually show.

No.	Verification Check	Evidence in Report	Status
1	The form accepts the required applicant details.	Fig. 7 shows the mandatory fields (Monthly Income, Loan Type, Consent For Credit Check) filled in.	**Shown**
2	The form submission is successful.	No submission confirmation screen was captured; successful submission is indicated indirectly by the email in Fig. 8.	**Indirect**
3	The application record is saved.	The saved record (report view) is not captured in the screenshots.	**Not shown**
4	The correct workflow is enabled.	Fig. 6 shows Loan_Application_Notification as Enabled with 1 Action.	**Shown**
5	The workflow contains the Send Email action.	Fig. 5 shows the Send mail action; Fig. 6 shows 1 Action.	**Shown**
6	The sender email address is verified.	Sender verification settings are not captured in the screenshots.	**Not shown**
7	The recipient field is mapped correctly.	Fig. 5 shows the action addressed to ${Email_Address}.	**Shown**
8	The acknowledgement email is received in the applicant's inbox.	Fig. 8 shows the email in the Inbox, received at 12:04 pm.	**Shown**

Status key: Shown = confirmed by a screenshot; Indirect = supported by other evidence; Not shown = not captured in the original document.

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8. Result

Thus, the workflow automation for the Bank Loan Application Form was configured successfully in Zoho Creator. The workflow was designed to execute when a new loan application is submitted successfully and to send an acknowledgement email to the applicant through the Send Email action.

The experiment demonstrates how workflow automation can simplify routine communication, reduce manual effort, improve the applicant experience, and provide timely confirmation of loan application receipt. The successful receipt of the acknowledgement email (Fig. 8) confirms the expected operation of the configured workflow.